

Chapter 6

Recursion and Prosodic Subordination

Abstract

There is an asymmetry in the assignment of prosody depending on linear order. The asymmetry is the following: when a functor precedes its complement, it can bear an accent, but when it follows its complement, it must be ‘prosodically subordinated’. This asymmetry is at the heart of the distribution of nuclear stress, and the phonological difference between the pre- and post-nuclear domain. A principle of PROSODIC SUBORDINATION is proposed that accounts for the asymmetry and derives the correct prosodic grouping and the distribution of accents, including the location of nuclear stress. It can be observed across different domains of the grammar, from the prosodic relation between affixes and stems to the relation between matrix clauses and embedded clauses, and it applies to the relation between heads and complements and to the relation between modifiers and modifiees.

6.1	Nuclear Stress and Prosodic Asymmetry	198
6.2	Prosodic Subordination	208
6.3	Nominal Arguments	212
6.4	Predicates and Prosodic Asymmetry	220
6.5	Further Evidence for Asymmetry	231

6.1 Nuclear Stress and Prosodic Asymmetry

The concept of ‘nuclear stress’ has played a crucial role in the development of theories of the syntax-phonology interface. This section discusses various problems with the original notion of nuclear stress and presents a systematic prosodic asymmetry. This asymmetry will be accounted for by the recursive application of PROSODIC SUBORDINATION. It will be shown to account for the prosodic grouping and nuclear stress in various syntactic domains of English, Dutch and German.

6.1.1 SPE

In SPE, for every phrase marker the nuclear stress rule (NSR) identifies exactly one vowel as most prominent. The vowel that native speakers perceive as most prominent in a given structure counts as its nuclear stress. The representation of stress in SPE is that of a phonological feature that can have any positive integer number as its value. The nuclear stress of an utterance is the unique vowel that has the value 1. Within every sub-constituent, there is exactly one vowel that has the lowest number. We can call this the vowel carrying nuclear stress within that sub-constituent.

A different notion of ‘nuclear stress’ or ‘nuclear accent’ is used in the intonation literature. Newman (1946, 174) singles out a heavy stresses as ‘nuclear’ if it ‘acts as the nucleus of an intonational unit’. Within any intonational unit it is always the last heavy stress that counts as the nuclear stress. For the course of this section, I will stick to the SPE notion of nuclear stress, and I will return to Newman’s idea about nuclear stress in the next section.

In SPE, just as in an earlier version of the theory presented in Chomsky et al. (1957), the stress value of a particular vowel is determined by recursively going through the structure and applying phonological rules. The three stress rules proposed for English in SPE are the Main Stress Rule (MSR), the Compound Rule (CR), or the Nuclear Stress Rule (NSR). The MSR assigns stress to mono-morphemic words and morphologically built complexes; the CR negotiates stress within compound structures; and the NSR finally negotiates stress in syntactic phrases.

These rules are recursively applied to surface structures, that is to the syntactic tree that is the output of a derivation. This basic idea of a recursive algorithm to assign stress was called the ‘transformational cycle’, and can be summarized as follows (SPE, 24–26):

“Utilizing the principle of the transformational cycle, the speaker of English can determine the phonetic shape of an utterance on the basis of such rules as the Compound and Nuclear Stress Rules, even though the particular utterances may be quite new to him. [...] Once the speaker has selected a sentence with a particular syntactic structure and certain lexical items (largely or completely unmarked for stress, as we shall see), the choice of stress contour is not a matter subject to further independent decision. [...] With marginal exceptions, the choice [...] is as completely determined as, for example, the degree of aspiration”.

Each of the stress rules promote one vowel over other vowels in prominence. This is achieved by ‘subordinating’ all other vowels in the structure by augmenting the stress number by 1. Which rule applies at a certain syntactic node depends on whether it is a word, a compound, or a phrase. In phrases, the NSR applies and demotes all stress numbers on each vowel except for the last vowel that carries a ‘1’.

The ultimate stress values of a phrase marker in English are determined by recursively applying the three stress rules to the tree structure of a phrase marker, starting at the terminal nodes:

- (1) The Nuclear Stress Rule in SPE

[_{NP} [_{DP} Helen] [_{VP} [_{V'} reads _V [_{NP} [_{N'} strange _A [_N books _N]]]]]				
1	1	1	1	MSR

		2	1	NSR
	2	3	1	NSR

2	3	4	1	NSR

This chapter is concerned with the type of data for which SPE would have considered the NSR to the relevant phonological rule, so I will not discuss the formulation of the MSR and the CSR. While the representation of prosody that is assumed will be used in this chapter is very different from the stress numbers in SPE, the notion of nuclear stress is still going to play a role, and the solution will again make use of the transformational cycle. The crucial difference will lie in the formulation of the principle that derives nuclear stress.

As for the NSR in SPE, it is sufficient for the present discussion to note that it has the effect that on any branching phrase, it is the last element that receives main prominence. In other words, nuclear stress is predicted to be always on the *rightmost element within phrases*. This prediction, however, turns out to be false.

6.1.2 Linear Order *vs.* Argument Structure

Newman (1946, 179) notes that in the following type of example, main prominence falls on the object, and not the final predicate¹:

- (2) a. fávör to ask.
b. bréad to eat.
c. instrúctions to leave.

Bresnan (1971) discusses this and related data to show that the theory of nuclear stress assignment in SPE is in need of revision. Bierwisch (1968, 176) points out

¹I only note nuclear stress for simplicity.

similar cases in German, and concludes that the rules of stress placement must make reference syntactic and semantic information:

- (3) a. Peter betrachtet ein Búch.
Peter looks.at a book
'Peter is looking at a book.'
- b. Peter hat ein Búch betrachtet.
Peter has a book looked.at
'Peter looked at a book.'

According to Newman, the crucial factor involved in what he calls 'middle stress', that is non-final nuclear stress, is argument structure: the 'logical object' precedes the predicate. Hence nuclear stress is final in the following examples, in which the 'logical object' follows the predicate:

- (4) a. a desire to éat
b. the will to líve
c. instructions to léave (...= was instructed to leave)

That predicates, and heads more generally, are less prominent than their complements holds across a number of languages, as illustrated in Donegan and Stampe (1983), Jacobs (1992) and Cinque (1993). In Germanic languages this is particularly easy to observe since they are languages with mixed headedness. In English, functors predominantly take their arguments to the right:

- (5) She painted some wálls.

But sometimes, the complement of a predicate can precede it. In English, in certain environments a predicate can follow its nominal complement. We can call these types of environments 'OV' environments.

- (6) She had some wálls painted.

Whether or not a predicate in fact subordinates depends on the meaning of the predicate, the meaning of the object, and the interaction between the two. For example,

while predicates in German OV environments tend to subordinate, sometimes they do not. This was observed in Höhle (1982) and Jacobs (1988, et. seq.). For example, in the following sentence it is equally possible to stress the predicate and the object:

- (7) a. Der Arzt wollte einen Pátienten untersúchen.
 the doctor wanted the patient examine
- b. Der Arzt wollte einen Pátienten untersuchen.
 the doctor wanted the patient examine

There does seem to be a subtle difference in meaning, but neither sentence requires narrow focus on the object or the predicate. The same is true for corresponding examples in English:

- (8) a. The dóctor wanted to have a pátient exámined.
- b. The dóctor wanted to have a pátient examined.

The difference has been related in the literature (e.g. in Krifka (1984) to the distinction between ‘thetic’ and ‘categorical’ statements—which is still very ill understood. I will return to this problem in section 6.3 and explore some of the differences between the two types of cases. For many cases, however, the predicate is subordinated and main stress falls on the argument.

The NSR in SPE predicts nuclear stress to be final in phrases, and thus cannot explain why nuclear stress falls on the object irrespective of linear order. Bresnan (1971) suggested to maintain the NSR proposed in SPE, and to derive the effect by changing the assumptions about how the phonological cycle interacts with the syntactic cycle. I will return to her proposal in section (6.3.3).

More recent approaches to sentence stress and prosody have incorporated Newman’s idea and make reference to argument structure in deriving prosody. The generalization these approaches try to capture can be stated as follows (adapting the formulation in Schmerling (1976, 82)):

- (9) Arguments are more prominent than functors.

There are different approaches to explain this observation. The first kind of approach involves the notion of ‘focus projection’. ‘Focus projection’ is what Höhle (1982)

termed the phenomenon that accents on certain constituents are compatible with focus on constituents that include them, sometimes up to and including the entire sentence. The NSR tries to capture the prosody of sentences with ‘neutral’ focus (a discussion of how this can be defined is given in chapter 7). Selkirk (1984) specifies explicit rules about which constituents can be in focus given a certain distribution of pitch accents. One of the rules of focus projection is that heads can inherit F-marking from arguments. This correctly captures that an accent on the object is sufficient to license ‘focus’ on the constituent containing both the head and the object, and allows the predicate thus to remain unaccented and hence lower in prominence. The approach in terms of ‘focus projection’ receives a detailed discussion in chapter 8.

The second type of approach attempts to derive the generalization from conditions on prosodic domain formation. Gussenhoven (1983) proposes a rule of accent formation, which integrates predicates into the accentual domain of adjacent arguments. Gussenhoven’s proposal is reminiscent of the approach taken in Truckenbrodt (1995), who introduces a constraint *Wrap-XP*, which forces XPs (but not heads) to be contained in a prosodic constituent of their own.²

The third type of approach explicitly postulates a principle that assures that predicates are less prominent than their arguments, as e.g. in Schwarzschild (1999). In Jacobs (1991a, 1992, 1999), Arregi (2002), Wagner (2002a) a version of this principle is incorporated into a theory that assumes the transformational cycle and assigns a complete metrical representation to phrase structures.³

A commonality of all of these approaches is that they treat the head-final case

²Approaches based on left- or right- XP alignment (Selkirk, 1986) and the approach based on ϕ -formation and restructuring in Nespor and Vogel (1986)—while not incompatible—offer no explanation for the generalization about complement prominence when a language has both head-initial and head-final constructions, and I will not discuss these approaches at this point. They do, however, correctly predict an asymmetry depending on the linear position of a head relative to its complement, to which I will turn shortly.

³Cinque (1993) proposes a different explanation for this generalization based on ‘depth of embedding’, and does not make direct reference to argument structure or the head-complement relation. This approach is discussed and criticized in Truckenbrodt (1993) and Arregi (2002).

and the head-initial case as mirror images of each other. The principles that assure that predicates are less prominent apply independent of linear order (e.g. Schmerling (1976, 82)). These symmetric approaches do not account for a systematic asymmetry that relates prosody and linear order.

6.1.3 Prosodic Asymmetry

The previous section illustrated that predicates are less prominent than their complements, whether they precede it or follow it. There is a crucial difference though between head-initial and head-final constructions: when the functor precedes its complement, it can optionally bear an accent, but if it follows its complement, it must be unaccented:⁴

- (10) Did she change anything in her new apartment?
- a. She páinted the wálls.
 - b. She had the wálls painted.

Newman (1946) observed that it is the last accent that is perceived as the more prominent accent. If this is correct, then placing an accent on the predicate shifts the prominence in the case when it *follows* the argument but not when it precedes it. The asymmetry observed here can thus be related to the more general observation that predicates are less prominent than their arguments in Schmerling (1976).

There are of course circumstances when the predicate can or even must receive an accent in even in the ‘OV’ order (11a).

- (11) What did she do with those ugly pink walls?
- She had the walls páinted.

⁴The context is such that the presupposition of uniqueness in the answer is fulfilled although the walls have not been referred to yet. ‘The walls’ is not contextually given. They are the walls pertaining to the apartment. The interaction between the presupposition of givenness marking and the presuppositions of definite noun phrases is discussed in chapter 7.

But this prosody requires the complement to be ‘given’ in the context. To be precise, the prosody in (11a) introduces the presupposition that that another proposition about ‘the walls’ is salient. This is the same presupposition that (12) introduces:

- (12) What did she do with those ugly pink walls?
 She painted the walls.

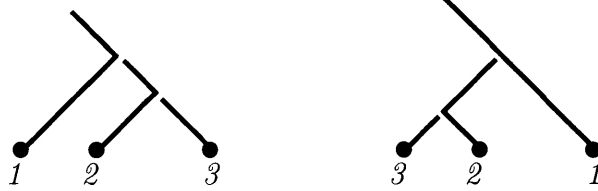
Whether or not an accent is placed on the predicate in the VO-order does not affect the information structure of the sentence. Gussenhoven (1983) reports experimental evidence that the presence/absence of placing an accent on the predicate has no influence on whether or not listeners are more likely to consider a sentence part of a dialogue in which the predicate was given or not.⁵

The presuppositions of givenness marking will be discussed in chapter 7. For the remainder of this chapter, I will only discuss sentences that have ‘neutral’ prosody, i.e. a prosody that does not impose a presupposition of ‘givenness’ on the context. The notion of ‘neutral’ prosody is difficult to define. It is made explicit in chapter 7.

The observation important here is that there is an asymmetry between VO and OV. This prosodic asymmetry can be further illustrated by looking at structures that consists of three elements that form a selectional sequence, such that the highest constituent Head_1 selects a complement that consists of a a head, Head_2 , and its complement. In the following, I will draw syntactic trees such that the projecting side of a branching node is not interrupted. There are several different linearizations for these structures. Consider the following two cases:

- (13) a. Right-Branching a. Left-Branching

⁵The experiment used a ‘context retrieval task’. Listeners had to choose between two dialogues, one in which the answer had wide focus, the other in which it had narrow focus (a narrow focus question makes the predicate given. Some researchers have claimed that accent on the predicate is infelicitous with narrow focus on the object (Kadmon, 2003)). But an experiment reported in Gussenhoven (1983) found that while there is at least a tendency to use unaccented predicates in contexts with narrow focus on the object, the correlation was not clear cut. Furthermore, listeners were not able to use the accentedness of the predicate to retrieve the context in which the answer was uttered.



Instances of both linear orders are attested in German predicate sequences. Consider the following two examples:

- (14) ‘...weil er ihr...
 ‘...because he...

- a. [versprách] [zu versúchen] [zu schweígen]. $\acute{1} \prec \acute{2} \prec \acute{3}$
 b. [zu schweígen versprach zu versuchen.] $\acute{3} \prec \acute{2} \prec \acute{1}$

be silent try promise

‘...promised her to try to be silent.’

The observation is that predicates that precede their complements receive a prosodic domain of their own, but predicates that follow their complement are prosodically subordinated. The metrical representation assigned to those structures by Jacobs (1991b) and Cinque (1993) is exactly symmetrical:

- (15) Symmetric Representations

- a. $1 \prec 2 \prec 3$ b. $3 \prec 2 \prec 1$

$$\begin{array}{cc} \begin{array}{c} \times \\ \left(\begin{array}{c} \cdot \quad \cdot \quad \times \\ \cdot \quad \left(\cdot \quad \times \right) \\ \times \left(\times \left(\times \right) \right) \end{array} \right) \end{array} & \begin{array}{c} \times \\ \left(\begin{array}{c} \times \quad \cdot \quad \cdot \\ \left(\times \right) \cdot \quad \cdot \\ \left(\left(\times \right) \times \right) \times \end{array} \right) \end{array} \end{array}$$

Cinque (1993, 245)

$\begin{array}{ccccc} \times & \times & \times & \times & \times \\ \times & & & \times & \\ & & & \times & \\ & & & (\times) & \end{array}$

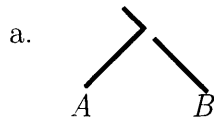
$\begin{array}{ccccc} \times & \times & \times & \times & \\ \times & & \times & & \\ \times & & & & \\ (\times) & & & & \end{array}$

Jacobs (1992)

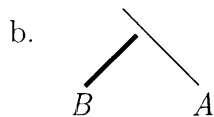
These symmetric approaches miss out on an important phonological generalization, namely that the two cases show a systematic prosodic asymmetry. The generalization about prosody and linear order can be summarized as follows.

In the following, I will draw trees with bold lines on the branches whose corresponding phonological representation do not undergo subordination; also, I am connecting the projecting side to the root and leave the non-projecting side unconnected.

(16) PROSODIC ASYMMETRY



When a functor *A precedes* its complement B, the functor may be on a par with its argument or may be prosodically subordinated: [Á B]; [A B].



When a functor *A follows* the complement B, A is prosodically subordinated: B A (unless A is focused or B is given).

In the next section, I will discuss how the effects of prosodic subordination can be accommodated with the cyclic approach to syntax-to-prosody mapping presented in chapter 7. In section (6.4) I will present more evidence for the generalization from predicate sequences in English, Dutch, and German. The generalization will be shown to be robust under word order variation across dialects of West-Germanic. Section (6.5) extends the generalization to a much wider variety of cases.

6.2 Prosodic Subordination

In theory of syntax-to-projection presented in chapter 2 prosodic subordination remains unaccounted. This section presents an addition to the theory that allows prosodic subordination to recursively apply.

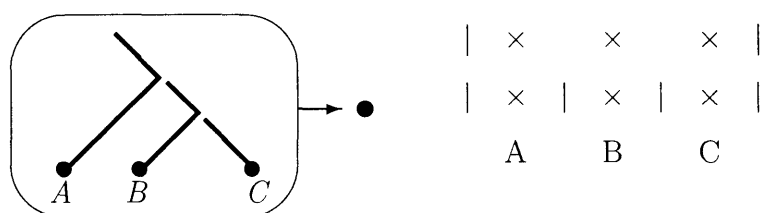
6.2.1 Concatenate and Project

Consider again the simple case of sequences of predicates in English. They appear to form a right branching structure, and they have a ‘flat’ prosody, in that each predicate receives an accent and is set off from adjacent predicates by boundaries of equal strength. They are prosodically on a par. Compare the prosody in (17a) with (17b), which is very similar in the prosodic phrasing:

- (17) a. They want | to try | to begin | to plan | to move.
 b. to want, | to try, | to begin, | to plan, | and to move.

In chapter 3 it was argued that both types of structures are associative domains, and are therefore assembled in a single cycle. Flat prosodies are generally associated with right-branching structures that are assembled in a single cycle, and are mapped to a single foot in the top metrical grid line. This is illustrated in the following diagram:

- (18) A or B or C: Derived in a Single Cycle



A single mapping principle was proposed in chapter 2 to derive prosodic structure: each cycle is mapped to a new top line grid line in the metrical representation upon completion. The mapping principle is the following:

(19) PROSODIC MATCHING

a. Concatenate

Concatenate the prosodic representation of the elements in the domain aligning their top lines and filling the columns where necessary.

b. Project

Create a new top-line grid line n by projecting all grid marks on line $n-1$, and mapping them into a single foot on line n .

This projection principle has the effect that each element in a cycle projects to the top line. That was the right result for coordination structures, since independent of the bracketing and the number of cycles, each conjunct always receives at least one accent.

Nuclear stress is not encoded here by projecting the element bearing nuclear stress to the top grid line, as is usually done (e.g. Halle and Vergnaud, 1987, Cinque, 1993, Arregi, 2002). Still, nuclear stress can be read off the structure. The generalization proposed in chapter 2 is the following:

(20) Nuclear Stress Generalization

Within each foot, nuclear stress is perceived on the last of those grid marks that project highest.

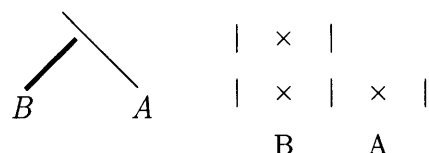
The idea that the most prominent is the last among equal goes back to Newman (1946), and has been adopted by various authors since (e.g. Schmerling (1976, 86) and Truckenbrodt (1995, 113)).

This means that nuclear stress is not a primitive of the theory but is a derivative notion. In order to determine the nuclear stress of a structure, one needs to investigate the conditions of accent omission and prosodic subordination.

While nuclear stress is always on the last element when all that happens is PROSODIC MATCHING, some times nuclear is not final, as was discussed above. For

example, in the case of the OV-word-order, something else happens: only some but not all material projects to the top line. I assume that the correct prosodic representation for prosodic subordination to be the following:

(21) Prosodic Subordination



Constituent A is ‘affixed’ to B, i.e. to the prosodic domain it is subordinated to. Since the A does not project to the top grid line, it does not receive an accent. All and only top line grid marks are associated with accents, as outlined in chapter 2. This has the effect that nuclear stress is not perceived on the final element A in the structure, but on B. *Nuclear stress is perceived on the last element that has not been subordinated.* But under what conditions does subordination apply?

6.2.2 Cyclic Subordination

The assumption in chapter 2 is that syntax is only mapped to prosody at the end of a cycle. The usual way of relating the prosody of two cycles is to concatenate a number of elements at the top line and projecting each elements. I will assume that subordinating one constituent to another constituent is a relation between two sister constituents, where one sister is subordinated and ‘affixed’ to the other. I will treat subordination then as a case in which two sister nodes undergo a cycle and the mapping principle that applies is PROSODIC SUBORDINATION. The following principle has the effect of subordinating one sister to the other.

(22) PROSODIC SUBORDINATION

A is subordinated to B by concatenating A and B such that if A projects to line i , B projects to line $i+1$.

The question is now under what circumstances PROSODIC SUBORDINATION applies. PROSODIC SUBORDINATION is the only principle apart from PROSODIC PROJECTION in the the system. All factors, be they related to argument structure or related to

information structure, factor into the derivation prosody only in deciding when either of the principles applies.

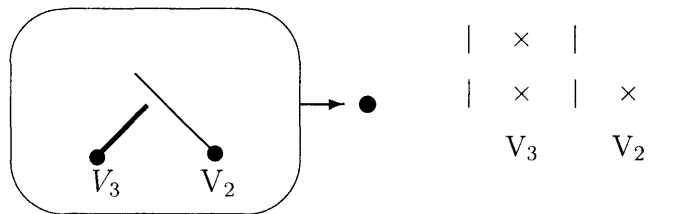
One factor that can trigger PROSODIC SUBORDINATION seems to be argument structure. The discussion in the previous section suggested that a functor follows its complement is obligatorily prosodically subordinated. We related this generalization to prominence: if the functor would not be subordinated, then it would project to the same grid line and become more prominent than its complement by virtue of the nuclear stress principle. The following principle is the driving force behind obligatory subordination:

(23) COMPLEMENT PROMINENCE

If A and B are sisters and A is the functor and B its argument, then B is more prominent than A.

This principle will force prosodic subordination whenever a functor follows its complement, due to generalization about prominence in sequences of unsubordinated constituents. If PROSODIC SUBORDINATION did not apply, then the functor would *not* be less prominent than its complement. The application of PROSODIC SUBORDINATION can be illustrated as follows:

(24) PROSODIC SUBORDINATION



The conditions under which PROSODIC SUBORDINATION applies in cases where the functor *precedes* its complement will not be discussed here. I will assume that it is optional for the time being, and indeed in many cases it is. Two factors seem to play a role, one is information structure, the other is rhythm. Some evidence for rhythmic effects will be discussed in passing in this chapter.

The following sections will apply the principle to a number of different types of cases and illustrate how the system works.

6.3 Nominal Arguments

This section discusses the prosody of complements and specifiers, and discusses issues relating to the interaction of movement and subordination.

6.3.1 Complements

Predicates preceding their complement can receive an accent in English (25a). This is true both for infinitival and for DP-complements (25b).⁶

- (25) a. She wanted to help to succeed.
b. She wanted to help to paint the house.

These sequences of predicates and their prosody can be derived by assembling them in a single cycle and applying PROSODIC PROJECTION upon spell-out, as outlined above. Some times, even in English an object precedes its predicate and the expected asymmetry is observed, as discussed before:

- (26) What did she want to change before moving in?
She wanted to have some walls painted.

⁶The accent on ‘paint’ is certainly not obligatory, but it is possible. In VO-orders, the verb *can* bear an accent. The evidence for the phrasing of verb and direct object often given is the application of the rhythm rule:

- (1) Evidence for Phrasing Kenesei and Vogel (1995)
a. ...in English: Rhythm Rule:
[They managed] [to outclass] [Delaware’s cantéen].
b. ...in German: No Accent on Verb
[Sie haben Delaware’s Kantine übertroffen].

Since here, the functor ‘painted’ follows its complement, ‘walls’, PROSODIC SUBORDINATION applies. Assuming that ‘some’ has already been subordinated relative to ‘walls’ at an earlier cycle, the following representation is derived:

- (27) some wálls painted
- | | | | |
|------|-------|---------|---|
| × | × | × | × |
| | | | |
| some | walls | painted | |

There are more cases where functors are preceded by arguments. Consider intransitive predicates and subjects: subordination is often the unmarked option, both with unaccusative (a) and unergative (b) verbs.

It has been reported that unaccusatives tend to phrase with the subjects whereas unergatives don’t (Selkirk, 1995b, Hoskins, 1996). This at least does not hold for *all* unergative and unaccusative predicates, as illustrated here. While it may well be that these subjects are not in the complement position of the lexical predicates involved, they are still the arguments of the the expression that follows them and they are not the projecting element. Therefore, they are the complement of the functors and the functors consequently subordinate. Subordination is even observed when there is more than one predicate (c):⁷

- (28) Why is everyone in the palace so nervous?
- a. [The quéen called]
 - b. [The quéen is expected to come.]

There are a number of cases that show that functors are not always subordinated when they follow an argument. As is well known, the subject in an intransitive sentence sometimes phrases separately from the verb, e.g. when it encodes information that

⁷The DPs here are new information definites. The presupposition of uniqueness is fulfilled by word knowledge: there is usually only one queen per palace, if there is a queen at all. This is why using an indefinite here would sound funny, since it suggests that the default assumption that there is only queen if any is not fulfilled. Definiteness is only very indirectly related to information status. See discussion in chapter 7.

counts as ‘given’ in the context. The following context is set up to facilitate wide focus in the embedded clause—but with a backgrounded or ‘given’ subject.

- (29) What did you say the dean did?
I just said that [The déan] [arrived].

This prosody is only felicitous if the argument in question is contextually given. An argument counts as contextually given if there is a proposition involving the same argument (or a co-referent one) that is salient in the discourse. This kind of effect and how it can be accounted for will be discussed in chapter 7. Other examples do not constitute discourse-old information. Consider the following:

- (30) Why did they close the factory?
a. [The fáctory] [went bánkrupt]
b. Gasolíne evaporated.
c. [A wórker] [eváporated].

‘The factory’ is given in (30a) and thus the verb receives main stress. ‘A worker’ in (30c) however does not have to be ‘given’ but still nuclear stress falls on the verb. The indefinite is arguably treated as a member of a set inferred from the background (the workers of the factory). Another factor may be that the predicate ‘evaporate’ is simply very unexpected in combination with the argument ‘a wórker’, as opposed to ‘gasoline in (b), for example. Just as in (a), the verb receives an independent accent.

Similar contrasts as in the case of unaccusative verbs can be constructed with unergative verbs. In (31b), the subject is again an indefinite that is interpreted as a member of set made salient by the discourse (i.e. a partitive relating to a set in the background).

- (31) Why did they interrupt the play?
a. [A chıld was crying].
b. [An áctor] [was crýing].

The stage-level, individual-level distinction is directly reflected in prosody (Schmerling (1976), Diesing (1992), Kratzer (1995)), and here discourse new-/oldness is not at stake. Individual-level predicates resist subordination:

- (32) a. [Your éyes are red].
 b. [Your éyes] [are blúe].

The issue that in argument-predicate sequence prosody is to some degree flexible and under ill-understood circumstances the predicate can also receive an accent does not only arise in the case of subjects. As observed in Höhle (1982), Jacobs (1988), Jacobs (1991b), Uhmman (1991), and Jacobs (1999), there are some object-verb sequences in which the predicates receives an accent, and it is not obvious that in those cases the object must is presented as ‘given’.⁸

- (33) a. Der Arzt wollte einen Pátienten untersúchen.
 the doctor wanted the patient examine
 b. Der Arzt wollte einen Pátienten untersuchen.
 the doctor wanted the patient examine

There is clearly a subtle difference in meaning, but it is hard to definite exactly the criteria that distinguish them. The hope is that once properly understood, it will turn out that depending on the prosody the functor/argument relations are affected and that this is why the prosody is different. I will not explore the issue further in this chapter.

6.3.2 Specifiers

A notorious problem for the assignment of prosody in general (cf. discussion in Cinque, 1993) and for the present proposal in particular is the prosody of specifiers. The following case presents another case of a two-place functor.

⁸The notion often invoked in order to describe the difference between those two sentence types is the distinction between ‘thetic’ and ‘categorical’ judgements (see e.g. Krifka (1984) and Lambrecht (1994, 137) for English and German). The problem with this distinction is that it has resisted a precise formalization and the difference remains hard to pin down (see Ladusaw (1994) for an overview).

(34) The wóman wore the hát.

Under the assumption that it is ‘wore the hát’ that takes the subject ‘the wóman’ as an argument (Marantz, 1984), the asymmetry in (16) would lead one to expect subordination of ‘wore the hát’ under ‘the wóman’—a patently false prediction of the system described so far.

One possible way out is to question the functor-argument relation. The subject might be introduced by a separate head, as discussed in chapter 3, following Schein (1993), Kratzer (1996), Pylkkänen (2002). But it may still be that this head is ‘affixed’ to the transitive verb. Schein (1993) e.g. suggests that agents are introduced by INFL, which at least morphologically is affixed to the verb. In this case, the problem persists, since still the subject is the argument of the following functor, the VP headed by the predicate.

The problem could be resolved only (i) by positing that it is the specifier which in fact projects, following the proposal in Starke (2001), who takes specifiers to be complement-taking projectors; (ii) by stipulating that the head that introduces the subject is really a suffix on the subject; or (iii) by following (Anagnostopoulou and Alexiadou, 2001) in positing that only one argument can remain the projection of a functor and exploiting the movement to of the subject to derive a relation between the subject and its sister that does not require subordination.

The generalization is that for transitive functors that take two arguments subordination is only possible between head and complement. Jacobs (1991a, 21) proposes to stipulate this as a restriction on subordination: a predicate can subordinate (in his terminology, the object ‘integrates’) if no other argument has integrated into the predicate. He notes that compound stress rule seems to capture a same generalization.

Arregi (2002) uses the definition of complement and specifier in Bare Phrase Structure (Chomsky, 1994) to capture the effect. A complement is the sister of a non-branching head; a specifier is the sister of a branching head. The generalization about prominence is then that if B is the head and A and B are sisters then A is prominent unless B is branching. This is precisely the compound rule proposed in

Libermann and Prince (1977).

For Arregi (2002), specifiers do not receive nuclear stress is due to the fact that the constituent they attach to is internally branching. The solution in terms of ‘branchingness’ requires some elaboration.

There are many cases where branching nodes *do* subordinate. One involves cases where an argument precedes several predicates:

- (35) Maria hat zu schweígen *versucht zu versprechen*.
Maria has to be silent tried to promise

Also, predicates that are compounds and are thus branching can subordinate:

- (36) Er wollte das Zímmer staubsaugen.
he wanted the room vacuum-clean
‘He wanted to vaccuum-clean the room.’

Only in cases where the non-projecting element is the second argument of a transitive functor does the principle of subordination fail to apply, but not in cases where the complement of an embedded predicate is the sister of a branching cluster of predicates. It seems that ‘branchingness’ has to be defined in such a way that it only counts other arguments of the heads in question in assessing branchingness. This is the generalization offered in Jacobs (1991a).

A similar problem arises in the case of coordination. Again, ‘and’ can be seen as a two-place function. The prosody is similar to the previous example:

- (37) Coordination
The wóman and the hát.

If we do not have a special condition to prevent specifiers from subordinating, the analysis that I would have to adopt in order to assign the right prosody is the following:

- (38)
-
- ```
graph TD
 DP1[DP] --- DP2[DP]
 DP1 --- DP3[DP]
 DP2 --- the1[the]
 DP2 --- woman[woman]
 DP3 --- and[and]
 DP3 --- the_hat[the hat]
```

This type of analysis was in fact proposed for coordination in Munn (2000). The first conjunct projects, and the second conjunct including coordinated is an adjunct to the first. But it is not clear whether this kind of analysis could be generalized to *all* specifiers.

In order to capture the facts regarding specifiers I incorporate the insights from Jacobs (1991a) and Arregi (2002) and modify the principle of COMPLEMENT PROMINENCE as follows:

(39) COMPLEMENT PROMINENCE

If A and B are sisters and A is the functor and B its argument, then B is more prominent than A unless A already contains an argument (i.e. is branching in the sense of BPR).

### 6.3.3 Subordination and Movement

Bresnan (1971) adopts the NSR proposed in SPE, i.e. NSR is always assigned to the right-most element within a phrasal domain, and advances a quite different type of solution to non-final nuclear stress. She relates cases like (40a), in which a predicate follows its argument, to cases like (40b), in which more material follows the direct object.

- (40) a. George has plans to leave.  
b. Helen left directions for George to follow.

The ingenious solution that Bresnan (1971) proposes to reconcile the data with the NSR proposed in SPE is that the NSR applies *before and after* the object move to the sentence initial position. This has the effect the direct object receives nuclear stress in its in-situ position, i.e. all other words get demoted by 1. This copy is then deleted and the moved copy is the only one in the structure with number 1. In other words, the stress numbers that are assigned are exactly those that are assigned to a structure in which the direct object is in-situ and receives nuclear stress. Once again,

argument structure is crucial—it is the base location of the object that is used to assign the correct stress, and it is in this position because it is the internal argument of the predicate.

(41) The Nuclear Stress Rule and Movement

|                                                                               |       |       |       |            |
|-------------------------------------------------------------------------------|-------|-------|-------|------------|
| [ <sub>NP</sub> directions [ <sub>S</sub> for George to follow directions. ]] |       |       |       |            |
| 1                                                                             | 1     | 1     | 1     | <i>MSR</i> |
| -----                                                                         | ----- | ----- | ----- |            |
|                                                                               | 2     | 2     | 1     | <i>NSR</i> |
|                                                                               | ----- |       |       |            |
|                                                                               |       |       | ∅     |            |
| 1                                                                             | 2     | 2     |       | <i>NSR</i> |
| -----                                                                         |       |       |       |            |

An approach that merely states that predicates are less prominent than arguments (e.g. Schmerling (e.g. 1976) cannot account for (40b), since the subject of the embedded clauses is also subordinated.

In Bresnan's system, non-final nuclear stress in phrases must necessarily involve movement, in other words all head-final structures that show nuclear stress on the non-head must be derived from underlying head-initial structures. This approach has the advantage that there is no reference to argument structure in stating the NSR itself.

The observation that predicates are less prominent than their complements even when they follow them has an interesting consequence in the Bresnan-system: all cases in which nuclear stress falls on the complement in an OV structure, the order must be derived from underlying VO,; more generally, each head-final structure with nuclear stress on the complement must be derived from an underlying head-initial order.

The present approach is compatible though with the effects of movement that are discussed in Bresnan (1971), although the derivation of the observed facts works differently. Consider the following representation for (40b):

(42) [ instructions [  $\lambda x$ . for George to follow x ] ]

The top node of the tree has two daughters, the DP ‘instructions’ and a one place predicate, ‘for George to follow x’. The DP is the complement of the  $\lambda$ -predicate, and consequently it is prosodically subordinated. The entire relative clause is the functor and it follows its complement.<sup>9</sup>

The principle of subordination can apply to syntactic nodes of any size. We will see other examples in which movement causes the subordination of remnant material, including the subordination of predicate sequences, of complex modifiers, of affixes, and of entire matrix sentences.

## 6.4 Predicates and Prosodic Asymmetry

This section presents evidence for the generalization in (16). Functors that follow their complement or part of it are prosodically subordinated.

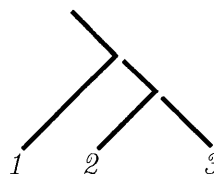
So far the main concern was data from English. I will now consider comparative data from different dialects of West Germanic. These languages differ in their prosody. However, once linear order is taken into account, most of the apparent prosodic differences (initial *vs.* final stress) reduce to syntactic differences (head-initial *vs.* head-final), and are as expected if the generalization in (16) holds.

### 6.4.1 One, Two, Three

Consider the case of predicate clusters in Dutch:

(43) Dutch Predicate Cluster: Final Stress

...[ dát hij ] [ wílde hélpen vérvén ].  
 that he wanted.to help.to paint




---

<sup>9</sup>It is crucial that the relative clause is counted as a one-place predicate, and it is not treated as branching for the purposes of COMPLEMENT PROMINENCE.



The predicates are ordered according to their embedding, starting with the highest predicate. The expectation is then that the prosody should be just like in the English case: a sequence of predicates that are prosodically on a par. The predicted prosodic structure looks as follows:

$$(44) \quad \begin{array}{c} | \quad \times \quad | \quad \times \quad | \quad \times \quad | \\ | \quad \times \quad | \quad \times \quad | \quad \times \quad | \\ \text{wilde} \quad \text{helpen} \quad \text{verven} \end{array}$$

The actual output for (43) usually contains fewer accents than given here: The accent on ‘helpen’ is usually dropped, as indicated in (45a). Another possibility is to only place an accent on the last predicate (45b):

- (45) a. ...[ dát hij ] [ wíldē helpen vérvēn]  
 b. ...[ dát hij ] [ wíldē helpen vérvēn]

Some predicates might be lacking accents due to rhythmic considerations. The distribution of accents on predicates seems to be sensitive to how closely adjacent accents are. This was observed also in Schmerling (1976, 98ff). This could be modeled by a phonological constraints forcing clash reduction that have the following effect:

(46) Clash Reduction

$$\begin{array}{c} | \quad \times \quad | \quad \times \quad | \quad \times \quad | \\ | \quad \times \quad | \quad \times \quad | \quad \times \quad | \\ \text{wilde} \quad \text{helpen} \quad \text{verven} \end{array} \rightarrow \begin{array}{c} | \quad \times \quad | \quad \times \quad | \quad \times \quad | \\ | \quad \times \quad | \quad \times \quad | \quad \times \quad | \\ \text{wilde} \quad \text{helpen} \quad \text{verven} \end{array}$$

One indication that rhythmic considerations play a role is example (47b). If a preposition separates the last two predicates, they are separated enough to both maintain their accents.

- (47) ...[ dát hij ] [ wíldē hélpen met vérvēn]

It seems that in a clash configuration some grid marks are deleted. The examples in (47a) and (47b) suggest that a relevant factor is whether two grid marks are separated by a foot on some lower grid line. Clash removal gets rid of the accent on ‘helpen’ in (45a) and on ‘met’ in (47):

- (48) Rhythmic Patterns  
 Clash Removal in ‘wilde helpen met verven’

|  |       |  |        |  |     |        |
|--|-------|--|--------|--|-----|--------|
|  | ∅     |  | ×      |  | ×   |        |
|  | ×     |  | ×      |  | ∅   |        |
|  | ×     |  | ×      |  | ×   |        |
|  | wilde |  | helpen |  | met |        |
|  |       |  |        |  |     | verven |

Clash Removal in ‘wilde helpen verven’

|  |       |  |        |  |        |  |
|--|-------|--|--------|--|--------|--|
|  | ×     |  | ∅      |  | ×      |  |
|  | ×     |  | ×      |  | ×      |  |
|  | wilde |  | helpen |  | verven |  |

The rhythmic nature of accent-placement in the pre-nuclear domain is further evidenced by (49a vs. b) and (49c vs. d) respectively.

- (49) Hij zēl dat hij...

- a. ...vērvde. í
- b. ...wil de vērven. 1<2  
 ...that he wanted to paint
- c. ...wilde helpen vērven. î<2<3  
 ...that he wanted to help to paint
- d. ...wilde kunnen helpen vērven. î<2<3<4  
 ..that he wanted to be able to help to paint
- e. ... wilde kunnen mōgen helpen vērven. î<2<3<4<5  
 ... wilde kunnen mogen helpen vērven. î<2<3<4<5

that he want be.allowed can help paint

‘He says that he wants to be allowed to be able to help to paint.’

It seems that in deciding which accent to maintain and which accent to remove, there are at least three factors to be considered: the last accent is never dropped; the first accent is not dropped, unless there are only two accents or *all* pre-nuclear accents are dropped<sup>10</sup>; medial accents are often all dropped, but if not then an alternating

<sup>10</sup>The literature reports a tendency to maintain accents at the beginning of domains (Bolinger, 1958).

pattern is chosen.

The following example illustrates that rhythm must be evaluated rather globally. In a matrix sentence in Dutch, the first predicate is considered for the computation of rhythm on the predicate sequence, although it has moved to first position and is not part of the cluster:

- (50) Hij wilde mogen            kunnen    helpen    vérvén.  
he   wants to be allowed to be able to help to paint.  
‘He wants to be allowed to be able to help to paint.’

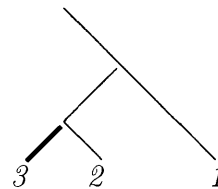
The reason I hesitate to formalize the rhythmic principles (which seem to require a rather global evaluation of eurythmy) is that it is not clear based on just those examples what the real pattern is. A closer investigation involving predicates of different shapes and sizes is necessary. This is a question for further research.

### 6.4.2 Three, Two, One

Consider now the German counterparts of the Dutch predicate clusters:

- (51) German Predicate Cluster: Initial Stress

...dass er málen helfen wòllte.  
...that he paint help want.



Main stress in German falls on the first predicate. No unreduced accents can occur in the post-nuclear domain unless one of the predicates is contrastively focused—in which case it would receive main prominence. The material following the accent on ‘malen’ has been severely pitch reduced if not deaccented. As indicated, there are intuitions about secondary stress in the post-nuclear domain.

German also differs from Dutch in the linear order of predicates, apart from the linear location of main word stress: while the predicates in Dutch are ordered ac-

cording to embedding, the order in German is the exact inverse. The distribution of accents is then as expected.

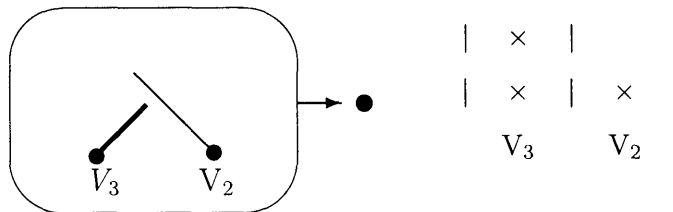
There are secondary stresses after the nuclear one, but these cannot be realized as pitch accents. This is true independent of the number of predicates that follow:

- (52) Er sagte dass er...
- a. ...málte.  $\acute{1}$
  - b. ...málen wollte..  $\acute{2} \prec 1$
  - c. ...málen helfen wòllte.  $\acute{3} \prec 2 \prec \grave{1}$
  - d. ...málen helfen können wòllte.  $\acute{4} \prec 3 \prec 2 \prec \grave{1}$
  - e. ...málen helfen kònnen dürfen wòllte.  $\acute{5} \prec 4 \prec \acute{3} \prec 2 \prec \grave{1}$
- that he want be.allowed can help paint

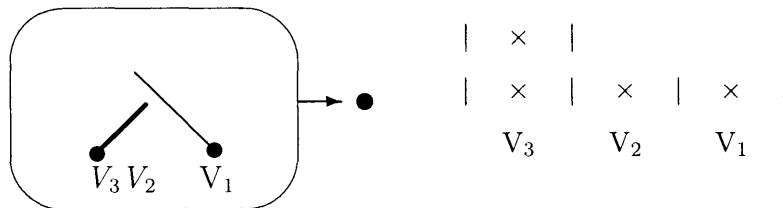
The fact that secondary stress in the tail of these clusters is rhythmic suggests that the prosodic structure in the tail is flat, and subject to rhythmic restructuring.

In order to derive the prosody in clusters with these linear orders, each predicate that is preceded by its complement must be subordinated. The derivation of (51) works then as follows, applying PROSODIC SUBORDINATION two times:

(53) First Cycle



(54) Second Cycle



The different word order compared to Dutch might be base-generated or one order might be derived by movement from the other—as long as movement involves  $\lambda$ -abstraction and no other operators it will leave the generalization in tact. Suppose the German order is derived from the Dutch order, then  $V_3$  is still an argument of its sister:

$$(55) \quad [V_3] [\lambda x. [V_2 x]]$$

If conversely the Dutch order is derived from the German order, then again, the functor-argument relations are not changed, only the word order is:

$$(56) \quad [[x V_2] \lambda x] [V_3]$$

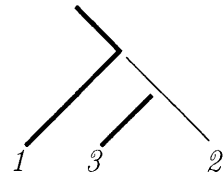
According to the theory presented here prosody does not give any evidence about which if any of the two orders is more basic.

### 6.4.3 Mixed Orders

That Dutch and German indeed do not differ in their prosodic systems becomes apparent in predicate cluster orderings that are attested in both languages:<sup>11</sup>

(57)

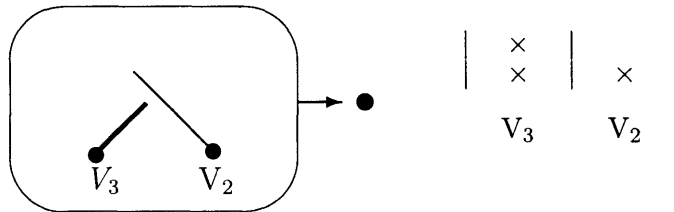
- a. ...dat Ján Maríe<sub>given</sub> kan<sub>1</sub> gezién<sub>3</sub> hebben<sub>2</sub>.  
that Jan Mary could seen have  
‘that Jan could have seen Mary.’
- b. ...wéil sie ihn hat<sub>1</sub> málen<sub>3</sub> wollen<sub>2</sub>.  
because she him has paint wanted  
‘that she has wanted to paint him’



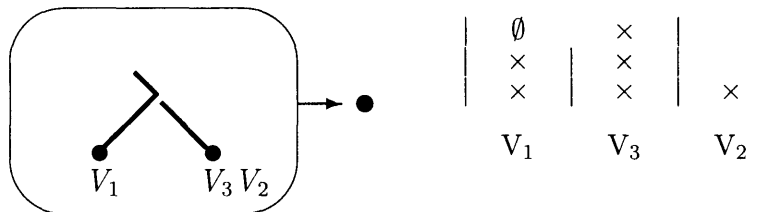
<sup>11</sup>Here, the DP argument preceding the cluster is made ‘given’ (old information), in order to prevent subordination of the cluster.

In this example, the second predicate ‘hebben’ is preceded by its complement ‘gezien’, which effects its subordination. The complete derivation involves a step of PROSODIC SUBORDINATION and a step of PROSODIC PROJECTION, followed by clash removal:

(58) First Cycle



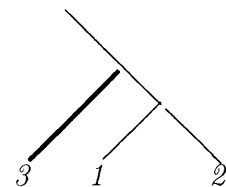
(59) Second Cycle and Clash Reduction



The next example illustrates a case where it seems that only a sub-constituent from the complement precedes one of the predicates (this word order is not accepted by all native speakers of Dutch, but is apparently generally accepted in Flemish). Predicate 1 is followed by predicate 2, but preceded by predicate 3, which semantically is part of its argument.

(60)

- a. ...dat Ján Maríe gezién<sub>3</sub> kan<sub>1</sub> hebben<sub>2</sub>.  
that Jan Marie seen can have  
‘that Jan can have seen Marie.’
- b. ...wéíl er es káufen<sub>3</sub> wird<sub>1</sub> können<sub>2</sub>.  
because he it buy will can  
‘because he will be able to buy it.’



This type of order can be derived by positing movement:

$$(61) \quad [V_3] [\lambda x. [V_1 V_2 x]]$$

The derivation proceeds then with a step of PROSODIC PROJECTION assigning a prosody to  $[\lambda x.V_1 V_2 x]$  creating a one-place functor, followed by PROSODIC SUBORDINATION. The derived representation looks as follows:

$$(62) \quad \text{Prosody for } 3 \prec 1 \prec 2$$

$$\begin{array}{ccccccc} | & \times & | & & & & \\ | & \times & | & \times & & \times & | \\ | & \times & | & \times & | & \times & | \\ V_3 & & V_1 & & V_2 & & \end{array}$$

The following cases of particle climbing further illustrate that entire predicate sequences can be treated as functors and be subordinated.

$$(63) \quad \text{Climbing up the Cluster (cf. Evers, 2001, and ref. therein)}$$

Het labyrinth waar we hem niet over...

- |    |                                              |               |                                                   |
|----|----------------------------------------------|---------------|---------------------------------------------------|
| a. | <u>zullen hoeven laten</u> <u>nā</u> denken. |               | $\dot{1} \prec 2 \prec 3 \prec \acute{5} \prec 4$ |
| b. | <u>zullen hoeven</u> <u>nā</u> laten denken. |               | $\dot{1} \prec 2 \prec \acute{5} \prec 3 \prec 4$ |
| c. | <u>nā</u> zullen hoeven laten denken.        | subordination | $\acute{5} \prec 1 \prec 2 \prec 3 \prec 4$       |
- about will need let think
- ‘The labyrinth about which we won’t let him reflect.’

I assume again that movement of the particle creates a one-place predicate:

$$(64) \quad [V_5] [\lambda x. [V_1 V_2 V_3 V_4 x]]$$

As expected the entire functor following the particle is prosodically subordinated.

#### 6.4.4 Predicate Sequences and DP Arguments

Subordination of predicates following arguments can also be observed in Dutch and German, even when multiple predicates follow an argument.

In Dutch, the entire cluster subordinates when the DP complement of the ‘deepest’ predicate precedes it, as it usually does. The prosody is exactly parallel to the case of particle preposing discussed above: the entire sequence of predicates is turned into a single functor that then takes the direct object as its complement:<sup>12</sup>

- (65) a. ...dat hij [een múur<sub>6</sub> wilde<sub>1</sub> mogen<sub>2</sub> kùnnen<sub>3</sub> helpen<sub>4</sub> vèrven<sub>5</sub>.]  
that he a wall want allow can help paint  
‘he says that he wants to be allowed to be able to help to paint a wall.’  
b. ...wéil er [ein Bild<sub>6</sub> malen helpen kònnen dürfen wòllte.]  
because he a picture paint help can be.allowed wanted

If the direct object encodes ‘given’ information, the predicates are not deaccented, however:

- (66) a. ...dat hij [de múur<sub>6</sub> wíldē<sub>1</sub> mogen<sub>2</sub> kúnnen<sub>3</sub> helpen<sub>4</sub> vérvēn<sub>5</sub>.]  
that he a wall want allow can help paint  
b. ...wéil er [das Bild<sub>6</sub> málen helpen können dürfen wollte.]  
that he the picture paint help able allowed wanted

‘...that he wants to be allowed to be able to help to paint a wall.’

This effect will be investigated in chapter 7.

---

<sup>12</sup>The post-nuclear stresses, both in Dutch and German, cannot receive an accentual realization unless the relevant predicate is focused or the preceding argument constitutes given information. A focused predicate, however, would receive nuclear stress. All predicates following the direct object are subordinated, that is, they are at least severely reduced in pitch range, if not altogether deaccented. The rhythmic pattern in the tail is the *same* in Dutch and German, despite the different word orders. This is as expected, since cyclic subordination derives a flat tail following the direct object for both languages.



## Discussion

Across the different languages, the prosodic generalization about predicate sequences is the same: the last accent falls on the complement that is selected last. This is captured here by the interaction of the principle of COMPLEMENT PROMINENCE with PROSODIC SUBORDINATION.

The generalization that nuclear stress falls on the most deeply embedded predicate can be captured applying various different approaches (Cinque, 1993, 269ff), Arregi (2002), or Jacobs (1991a, 1992)). The prosodic asymmetry in (16) discussed here however is not accounted for in those earlier approaches.

The generalization seems to be the same in English, Dutch, or German. While these languages differ in the precise tonology that is mapped onto the grid structures, the metrical grid is derived by the identical mechanism. The variation lies in the word order.

There is more evidence that the prosodic systems of those languages are identical. We can replicate all the possible orders of predicates across dialects in a *single* language. The following paradigm shows three of the possible orders of a particular predicate sequence in German. When predicates are ordered according to embedding as in (a), this order is often taken to involve ‘extraposition’. Different orders are possible, however, in so-called ‘restructuring’ environments (e.g. in b,c).<sup>13</sup>

- (67) ‘...weil er ihr...  
 ‘...because he...
- |                                               |                                             |
|-----------------------------------------------|---------------------------------------------|
| a. [versprách] [zu versúchen] [zu schwéigen]. | $\acute{1} \prec \acute{2} \prec \acute{3}$ |
| b. [versprách] [zu schwéigen zu versuchen].   | $\acute{1} \prec \acute{3} \prec 2$         |
| c. [zu schwéigen zu versuchen versprach.]     | $\acute{3} \prec 2 \prec 1$                 |
| d. [zu schwéigen versprach zu versuchen.]     | $\acute{3} \prec 2 \prec 1$                 |

---

<sup>13</sup>There are many syntactic differences between ‘extraposition’ and ‘restructuring’ that I will not address in this paper—restructuring derives what appear to be mono-clausal constructions that, e.g., facilitate scrambling between clauses, and allow pronouns that are arguments in the lower clause to be affixed on the matrix verb in second position, etc. (cf. Wurmbrand, 2003). Restructuring does not always result in a different word order between the predicates (Haider, 1994).

be silent try promise

‘...promised her to try to be silent.’

The example (67a) is similar to predicate clusters in Dutch (49), in that main stress is rightmost and secondary accents precede the main one. The fact that the median predicate does not necessarily lose its accent rhythmically as in the Dutch example (43) (although it may in fast speech), maybe due to the fact that there is unstressed phonological material—the preposition—intervening, preventing a clash. This is similar to the pattern observed in the Dutch example in (47), where also a preposition separated two predicates.<sup>14</sup>

Bech (1955/57) discusses stress and syntax of predicate clusters in a dialect that allows the following:<sup>15</sup>

(68) Unacceptable in Standard German, yet speakers have intuitions about main stress...

a. ??man wird ihn hier können liegen bleiben lassen.

b. ??dass man ihn kann lassen liegen bleiben.

c. ??dass man ihn hier wird können lassen liegen bleiben

that one him here will can let lie stay

‘that one will be able to let him stay lying here.’

---

<sup>14</sup>Since English does not allow cluster formation and has a rather fixed word order between predicates, only the evidence from DP’s preceding predicates discussed in the next subsection serves to show subordination in this language. A look at complex nominals, however, shows further evidence for the asymmetry at work:

(1) a. [ a tráiner ] [ of sláyers ] [ of vámpires ]

b. [ a tráiner ] [ of vámpire slayers ]

c. [ a vámpire slayer trainer. ]

<sup>15</sup>Compare: Voicing assimilation in English plurals is not segment-based: Speakers know that the plural of Bach is Bachs. (Morris Halle) Thanks for David Pesetsky for pointing out the parallel in the argument.

Speakers of Standard German know a syntactic generalization about stress—not a dialect-specific stress rule. The evidence discussed so far in this chapter suggests that the grammar of English and Dutch involves the exact same generalization about prosody.

The system has nothing to offer in order to explain *how the different linear orders come about*. This is a research question that is the topic of much current debate, and I will not endeavor to explore the issues here (for discussion see the collection of articles in Kiss and Riemsdijk (2004)).

## 6.5 Further Evidence for Asymmetry

This section presents further cases of prosodic subordination. The uniting factor is that they involve functors taking arguments, and the functors are spelled out before the argument.

### 6.5.1 Modifiers

Modifiers in Germanic differ from arguments with respect to their prosody relative to a following predicate (Gussenhoven, 1984, Krifka, 1984, Selkirk, 1984, Jacobs, 1992, Truckenbrodt, 1993). Consider first the case of arguments (the relevant predicates are in italics, the relevant arguments are underlined).

(69) Arguments

- a. [ Sie hát ] [ einen Tángo *getanzt.* ]  
           she has     a.acc Tango danced  
           ‘She danced a tango.’
- b. [ Sie íst ] [ in den Gärten     *getanzt.* ]  
           she is       in the.acc garden danced  
           ‘She danced into the garden.’

Predicates following DP-arguments (69a) and PP-arguments (69b) are subordinated. Subordination means that they are realized with a reduced pitch range, if not alto-

gether deaccented. They are prosodically ‘affixed’ to the preceding domain.<sup>16</sup> Consider now the case of modifiers (again, I put the functor in italics, and I underline the argument of the modifier, i.e. the modified constituent):

(70) Modifiers

- a. [ Sie hát ] [ *den ganzen Abend* ] [ getánzt. ]  
       she has    the.acc entire evening   danced  
       ‘She danced the entire evening.’
- b. [ Sie hát ] [ *im Gärten* ] [ getánzt. ]  
       she has    in.dat garden       danced  
       ‘She danced in the garden’.

Modifiers in (70) are mapped to their own accentual domain, separate from the accentual domain of the verb.<sup>17</sup> The difference between modifiers and arguments outlined so far is only one half of the pattern, however.

A closer look at the prosody of modifiers reveals an asymmetric pattern depending on linear order—in fact, a pattern that is exactly parallel to the asymmetry observed between predicates and arguments (Wagner, 2004a).

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<sup>16</sup>One could say that the arguments ‘incorporate’ into the prosodic domain of the verb that selects them; or conversely, that the verb becomes affixed to the prosodic domain of the argument. Indeed, structures that involve actual incorporation or affixation in German show similar prosodic patterns. I will not explore this point further in this paper.

<sup>17</sup>The following examples were presented in Cinque (1993, 250, 254) with nuclear stress on the adjunct—this, however, is not the appropriate prosody in neutral contexts. Unless the modifier is focused, nuclear stress in these examples falls on the verb, as was pointed out already in Truckenbrodt (1993), Jacobs (1999), and Ishihara (2001).

- (1) a. ...[dass Kárl] [ein Búch] [mit Múehe] [lésen kann].  
       that   Karl a   book with problems read
- b. ...[dass Frítz] [gút] [kóchen kann].  
       that   Fritz well cook   can.

The word order in (1) actually seems to require marked focus on the predicate. The more neutral order is: ‘dass Karl mit Múehe ein Búch lesen kann’. The example may then actually not be relevant for establishing the neutral stress pattern, since both word order and focus structure are not neutral.

When a modifier precedes the modified constituent, two separate prosodic constituents are derived, as was illustrated in (70); however, when it follows the modified constituent, the modifier is subordinated. Consider as a first example the case of post-posed sentential modifiers:

(71) Postposed Adjuncts

- a. [ Sie hát ] [ getánzt, *den ganzen Abend* ].  
           she has     danced   the.acc entire evening  
           ‘Sie hat danced the entire evening.’
- b. [ Sie hát ] [ getánzt, *im Garten* ].  
           she has     danced   in.dat garden  
           ‘She danced in the garden’.

The prosodic relation between modifiers and modified constituent is similar to that between predicates and arguments. Earlier treatments (e.g. Krifka, 1984, Gussenhoven, 1983, Selkirk, 1984, Jacobs, 1992) focused on the contrast between modifiers and arguments, and thus missed out on the parallel between modifiers and predicates.<sup>18</sup> More evidence for prosodic asymmetry in modification is presented in Wagner (2004a).

The prosodic parallel between modifiers and predicates can be linked to a semantic parallel: it is the modifier that is the functor and that takes a complement, the modifiee (Bierwisch, 2003). But if modifiers are the functor, then ‘Functor First’ requires them to be spelled out before the modifiee.

Modifiers that precede their complement can also be subordinated and remain unaccented, just like predicates in VO configurations. Again, whether or not they are depends on rhythmic factors:

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<sup>18</sup>It is not obvious whether it is legitimate to call a phrase like ‘the entire evening’ a modifier, and group it with complement taking predicates, since maximal projections are not usually taken to be able to take complements. Starke (2001) however argues for treating entire specifiers as projecting the feature of a functional projection and thus being the projector. Another way to make sense of this is to adopt the analysis of adverbials proposed in Alexiadou (1997) and Cinque (1999), who view adverbials as specifiers of functional projections. It is not ‘the entire evening’ itself then, which is the modifier, but a functional head that takes the DP as its specifier.

In (72a), it is possible to omit the accent on the manner adverb ‘gut’ in (72a) without changing the information structure of the sentence. This seems harder in the case of a longer adverb, that places more unstressed material between its main stress and the following verb (72b).

- (72) Sie gláubt, dass er  
 she believes that he
- a. [ gut kóchen kann].  
 well cook can.
- b. [ áusgezeichnet ] [ kóchen kann. ]  
 excellent cook can.

Selkirk (1984, 231) states that modifiers can be stressed without affecting the information structure. The example used also involved a modifier that *precedes* its modifiee:

- (73) Jóhn bought a red tíe.

The expectation based on the proposal here is that modifiers that precede the modifiee can but do not have to be accented, while modifiers that follow their complement must be deaccented.

Some modifiers do not seem to obey the generalization. They are accented although they follow what they modify. But there is reason to believe that the modifiee *is not their complement*. They differ in their syntactic relation to the modifiee, and have a structure similar to the relation between a secondary predicate and its argument.

An instructive case to look at are locatives. Locatives have different readings, again depending on where in the structure the locative is placed. Consider the following wide-scope locative, which Maienborn (2001) characterizes as setting the frame for the proposition expressed by the rest of the sentence. The presence of an additional sentential locative modifier forces the frame-setting reading (74), since two locatives of the same type would be contradictory:

- (74) In Italy, Lothar bought his suits in France. (Maienborn, 2001, 197)

The intended reading is one where the frame setting locatives narrows the time under consideration to ‘the time when Lothar was living in Italy’, while the lower locatives specifies where the events of buying suits took place. There are two possible linearizations of frame-setting locatives, and again the prosodic asymmetry is observed:<sup>19</sup>

- (75) a. [ In Italy, ] [ Lóthar bought his suits in France ].  
 b. [ [ Lóthar bought his suits in France ] in Italy ].

Frame-setting locatives contrast with lower locatives such as ‘in France’ in the examples above, which receive an accent sentence finally . Another difference between the two types of locatives is that lower Locatives cannot occur in sentence-initial position:

- (76) a. \*[ [ In France, ] [ Lóthar bought his suits ] in Italy ].  
 b. \*[ [ In Italy ] [ In France, ] [ Lóthar bought his suits ] ].

Frame setting locatives pattern just as is expected of functors: when they follow their complement, they are subordinated, but not (necessarily) when they precede their complement.

Low Locatives do not show the pattern expected of functors. But VP-final modifiers also differ in their syntax, and there is good reason to assume that they are not functors that take the VP as their complement. Larson (2005) argues that they are event predicates, that contain an event variable, and assigns a right-decending structure, just as in coordination structures. If Larson’s coordination analysis is correct, then the accentedness of low locatives is expected. This point is discussed in detail in chapter 3.

A final issue that again arises both for predicates and for modifiers is that of specifiers. Just as in the case of transitive predicates, an issue arises with transitive modifiers. Consider the nominal adjuncts involving ‘with’. ‘With’ would have to be analyzed as a modifier taking two arguments:

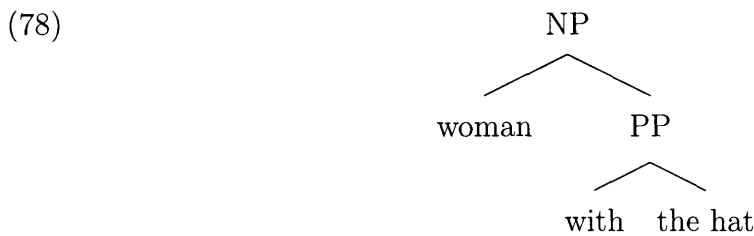
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<sup>19</sup>Jacobs (2001) and Maienborn (2001) note that frame-setting modifiers are similar in syntactic position and in their semantics to topics. The same parallel may apply to the intonational realization of these modifiers and topics. I will not explore this in any detail here.

(77) The wóman with the hát.

Given the analysis so far, the modifier should be the main projection of the construction. ‘with’ combines with ‘the hat’ forming the modifier ‘with the hat’, modifying the head noun ‘woman’. We would expect then subordination of ‘with the hat’, since it follows the modifiee—a prediction that is quite apparently false. Why does the principle of subordination fail to apply?

Note that even if ‘with’ is the main predicate, the entire constituent is not a PP, but an NP. The determiner ‘the’ selects a noun, not a prepositional phrase.



Of course, if this is the correct structure and the projector is really the head noun ‘woman’, the problem with respect to the prosody disappear: the projector precedes the non-projector, so two accentual domains are expected.

While ‘woman’ is the argument of the functor ‘with the hat’, the functor remains unsubordinated and receives its own accentual domain since it is its sister that projects.

### 6.5.2 Affixes

The principle ‘Functor First’ is also at play when syntax combines morphemes with each other. There is a well-known asymmetry between prefixes and suffixes which is observed across the Germanic languages but also attested cross-linguistically (e.g. Cohn, 1989, for Indonesian). Prefixes can form a prosodic domain of their own, while suffixes prosodically subordinate, i.e. they are realized with an extremely reduced pitch range if they carry accents at all:

(79) Stress: Prefixes vs. Suffixes

a. [ prè ] [ émptive ] *vs.* [ [ trúst ] wòrthy ]



- b. [ nòn ] [ hýper ] [ artìculate ] *vs.* [ [ [ fríend ] ship ] wòrthy ]

In Germanic, there is a correlation that suffixes tend to encode heads that determine the syntactic category while prefixes are modifying heads that leave the category of the stem unchanged. Since both modifiers and derivational heads can be conceived of as functors, this difference is not relevant for the prosodic facts we are interested in.

Just as with other functors, e.g. those entering predicate clusters, suffixes can be recursively added, and each of them is prosodically subordinated:

(80) Recursive Affixation

|                                               |                                            |
|-----------------------------------------------|--------------------------------------------|
| <u>kost</u>                                   | stem                                       |
| <u>kost</u>   <u>bar</u>                      | ‘expensive’                                |
| <u>kost</u>   <u>barkeit</u>                  | ‘valuable object’                          |
| <u>kost</u>   <u>barbkeitslos</u>             | ‘without a valuable object’                |
| <u>kost</u>   <u>barkeitslosigkeit</u>        | ‘state of being without a valuable object’ |
| <u>kost</u>   <u>barkeitslosigkeitshalber</u> | ‘because of a state of being without—’     |

The rhythmic distribution of intuitions about secondary prominence in the tail of derivatives suggest that the suffixes are on a par with respect to each other, just as is the case in predicate clusters.

### 6.5.3 VP-Preposing

The asymmetry presented so far relates prosody and linear order: if a functor *precedes* its complement it may but does not have to subordinate; if it *follows* its complement it is obligatorily subordinated. The asymmetry was first illustrated based on predicates and infinitival and nominal complements; it has been illustrated in a variety of other cases, including function words, instances of modification, and affixation.

A question not touched upon in this chapter so far is how the different linear orders between functors and their complements come about. Under an asymmetric view of syntax (e.g. Haider, 1993, Kayne, 1994) one order is basic and the other order is derived. Consider the following prosodic pattern of VP remnant fronting in German:

- (81) a. [ Péter [ hat der María ] [ einen Kúss gegeben ]  
           Peter   has the Mary     a kiss given
- b. [ [ Einen Kúss gegeben ] hat Peter der Maria. ]  
      a kiss given                   has Peter the Mary
- c. [ [ Einen Kúss gegeben ] hat Peter der [ Mariía]<sub>Focus</sub> ]  
      a kiss given                   has Peter the   Mary

The fronted VP receives an accentual domain, the remaining part of the sentence is subordinated—unless there is a focused constituent in the remnant material. This, of course, is exactly the asymmetry observed earlier for predication and modification.

It seems tempting then to analyze the reorderings at stake in all examples as cases of movement. A comprehensive syntactic account of the derivation of linear order in tandem with prosody remains to be developed.

#### 6.5.4 Matrix Clauses

The asymmetry can be observed for functors and complements of varying size. Just as it is attested in the combination of morphemes, it is also attested in the combination of sentences. In order to illustrate this point, I will use examples from German.

A matrix clause can either precede or follow an embedded clause. When it precedes it, both clauses are mapped to separate prosodic feet that are on a par:

- (82) [ Anna sagte zu Ulli ] [ wahrscheinlich werde es regnen ]  
       anna said to ulli     probably           will   it rain  
       ‘Anna said to Ulli that it will probably rain.’

However, when the matrix clause follows its complement, it is prosodically subordinated. (I mark the subordinated part by underlining):

- (83) [ wahrscheinlich werde es regnen ] sagte Anna zu Ulli ]  
       probably           will   it rain     said anna to ulli  
       ‘Anna said to Ulli that it will probably rain.’

Matrix clauses can be recursively nested:

- (84) [ wahrscheinlich werde es regnen ] meine Hans behauptete Anna ]  
           probably           will   it rain       thinks Hans claimed Anna  
           ‘Anna claimed that Hans thinks it will rain tomorrow.’

How can we account for this pattern? A matrix clause contains a predicate that takes an embedded clause as its complement. An important fact is that complement clauses obligatorily extrapose in German. This can be shown looking at cases where there is a clause final participle:

- (85) a.       Hans hat gesagt, dass es morgen   regne.  
               Hans has said     that it tomorrow rains  
       b.       \*Hans hat dass es morgen   regne gesagt.  
               Hans has that is tomorrow rain   said

Matrix clauses can then be viewed as functors and embedded clauses as their arguments once extraposition has applied:

- (86)    $\lambda x$ . Peter said x (   
           [ It will rain tomorrow. ] )

Since the matrix clause is the functor, and the embedded clause the argument, the linear asymmetry is now just an instance of the more general asymmetry described by (16). For a discussion of the syntactic and prosodic properties of these reorderings see Wagner (2004b).

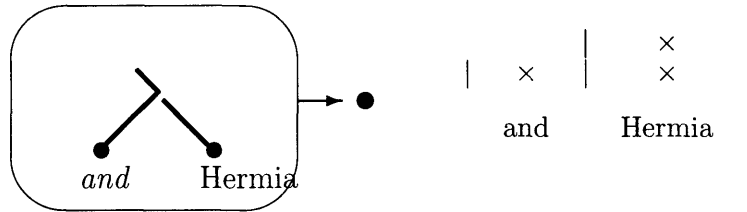
### 6.5.5 Function Words

In the discussion of coordination in chapter 2 it was left open how to account for the prosody of the connectives ‘and’ and ‘or’. Each conjunct receives an accent, but connectors usually do not:

- (87)   Lysánder and Hérmiá.

The subordination of function words is just another instance of the subordination of functors more generally. What is particular with function words is that usually, subordination takes place even when the functor precedes the argument. This is optionally possible with *any* functor.

(88) a. First Cycle



Prepositions can quite generally subordinate and remain unaccented:

- (89) a. of Demétrius  
 b. from Áthens  
 c. has lived

But note that the subordination of prepositions depends on syllable count, suggesting that this again a rhythmic effect, while post-positions obligatorily subordinate:

- (90) a. entláng des Mississíppis  
 along.location the Mississippi  
 b. # den Mississíppi entláng  
 the Mississippi along.direction

Adpositions thus give more evidence for prosodic asymmetry. Some function words have completely stressless allomorphs, e.g. 'v, ans 's for 'have' and 'is'. The hypothesis that I want to advance here is that function words can be only be subordinated or stressless when they occur in a position that *is independently prosodically subordinated*. In other words, there is nothing special about function words—they simply occur in weak positions, just like other functors. Stressless allomorphs are special in their underlying form, but they are not a different grammatical category.

If this is the correct account of the subordination of function words, then only functors that take complements should reduce in this way. Intransitive prepositions (Emonds, 1976) indeed do not subordinate (thanks to Draga Zec for suggesting intransitive prepositions as an example). This prediction is borne out:<sup>20</sup>

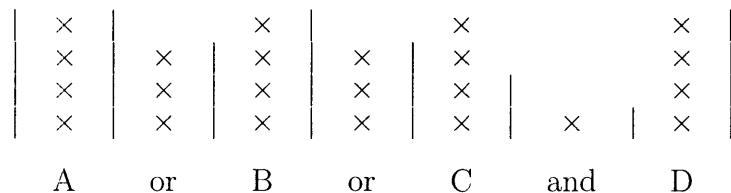
<sup>20</sup>A similar contrast exists with focus particles. While 'only' may subordinate, 'too' does not and is accented:

- (91) a. He wént awáy.  
 b. She lóoked aróund.  
 c. She wálked ín.

However, there are function words, or at least allomorphs of function words, that can never be accented or receive any prominence. I will assume that these are lexical allomorphs that are marked as prosodically deficient.<sup>21</sup> One possibility is to assume that these are lexical allomorphs specified to occur only in weak positions. These allomorphs develop for some lexical functors but not others because they occur more frequently.

Another fact that we can explain now is why the prosodic ‘size’ of functors should depend on their level of embedding, a result from chapter 5. Consider the following structure. The prosodies of the connectors i differ in at what grid line they form a foot. The ‘or’s form feet at a higher line compared to the ‘and’.

- (92) A or B or (C and D)



In chapter 5 I presented evidence that this is reflected in their phonetic realization: the degree of lengthening correlates with the strength of the upcoming boundary. Function words are prosodically subordinated *to the next lower grid line*.

- 
- (1) a. Máry only láughed.  
 b. Máry laughed, tóo.

But here, it is less obvious what the syntactic struture at least of ‘too’ is.

<sup>21</sup>There are also function words that lean to the ‘wrong’ side.

- (1) a. a friend’v Jóhn  
 b. John’s in Berlín.

Many approaches to function words, e.g. Selkirk (1996), make function word specific constraints. The implementation works by positing alignment constraints that make reference to the distinction lexical word only (*vs.* function word). Since function words are invisible to alignment constraints, they end up dangling on to the prosodic constituents they are adjacent to:

(93) (fnc ( lex )<sub>ω</sub>)<sub>φ</sub>

The class of function words is assumed to be defined based on independent syntactic grounds. The account does not explain though why it is that only function words subordinate that have a complement. One factor could be that they are phrase final. But many function words do subordinate phrase finally:

- (94) a. What are you *tálking* about?  
       b. Who are you *thínking* of?

Phrase finality does have an effect here though: phonetically reduced versions of function words are often not allowed when the complement is extracted, even when the preposition subordinates (see Selkirk (1996) and references for evidence):

(95) \*Who are you thinking'v

But still, the fact remains though that intransitive but not transitive functors subordinate as illustrated by (94) *vs.* (91) , and this fact receives a straightforward explanation within the allomorphy approach proposed here, under which the use of a reduced function words is parasitic on the it being independently subordinated.

Another issue is the notion 'function word' itself. It is not clear that there is a definition of function words that separates them from other functors in such a way that correlates with prosodic behavior. One criterion offered in Abney (1987) for function-wordhood is the open class/closed class distinction. But this distinction is not very helpful in predicting prosody. Prepositions are a closed class, and lexical verbs are an open class. But lexical verbs with the same or nearly the same segmental content have exactly the same prosody as prepositions:

- (96) a. hinder den Tórmann!  
           block the goalie  
           ‘Block the goalie!’
- b. hinter den Tórmann!  
           behind.direction the goalie  
           ‘Behind the goalie!’

Some prepositions (of, for...?), namely those that have prosodically sharply reduced allomorphs seem to be function words and are completely stressless, but others (along, behind...) don’t seem to be any different from lexical functors. Another criterion for assigning a morpheme to the class of function words that Abney (1987) discusses is prosodic reduction—which of course we cannot use here. Another less clearly defined notion is lack of lexical content. Unless there is any *syntactic* definition of exactly function words that are *prosodically* special, the theory is circular.

It seems more adequate then to treat Function words just as any other functor, but stipulate that those functors that occur in subordinated positions (i.e. those that take complements) can develop stressless allomorphs that are obligatorily stressless.

Which function words develop those allomorphs might relate to frequency, or *probability* as in (Jurafsky et al., 2000). If the idea that frequency or probability is relevant is correct, this might explain why it seems that whether or not words have *alternatives* in a given slot matters (over *vs.* under *vs* on...). In many contexts, there are few or no alternatives, and thus the probability of what can occur in those slots is very high (auxiliary positions, instances of ‘of’ and ‘to’). There is thus a potential explanation for the special status of some functors that does not require making a syntactic distinction between function word and lexical word.

A look at more sets of facts involving function words e.g. the one discussed in Selkirk (1996) and Zec (2002) would be useful at this point. The intricacies are mostly related to the fact that function words can be stressless and be incorporated into the prosodic word domain of their host. I will not discuss word morphology in any detail, and leave this for some other occasion.

The main points of this section were that (i) function words are subordinated just were other functors are and show the same asymmetry depending on linear order;

(ii) only functors that take a complement have weak allomorphs, but not intransitive ones—these are the ones that do not undergo subordination since they are the complement of a predicate; (iii) weak allomorphs of functors cannot occur in positions that are not independently subordinated.

## Summary

This chapter presented a generalization about prosodic asymmetry. Functors that follow their complement are obligatorily prosodically subordinated, but functors that precede their complement do not have to be.

The asymmetry can be captured by the interaction of syntactic recursion with two principles of syntax-to-phonology mapping, PROSODIC MATCHING and PROSODIC SUBORDINATION.

A look at predicate sequences illustrated that apparent prosodic differences between English, Dutch, and German reduce to independently motivated syntactic differences. The analysis can be applied to a variety of constructions involving functors of different kind and size.

The Cyclic application of PROSODIC MATCHING and PROSODIC SUBORDINATION derive nuclear stress: the last unsubordinated constituent counts as the nuclear stress. This is an architectural difference to alternative approaches to nuclear stress who usually first assign nuclear stress and then specify special rules to capture the difference between the post-nuclear and the pre-nuclear domain. The advantage of the approach in terms subordinating and then deriving nuclear stress indirectly is that it lends itself easily to incorporate information structural effects that can also determine the choice between PROSODIC MATCHING and PROSODIC SUBORDINATION.